

DONGHAE, Korea, Republic ofWMO: **471060**Lat: **37.5069N**Lon: **129.1244E**Elev: **136**StdP: **14.62**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	22.4	-13.0	2.7	23.3	-8.6	3.5	26.7	24.0	29.4	20.1	30.1	8.9	270	0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	87.9	75.0	85.2	74.2	82.9	73.5	77.9	83.8	76.7	82.5	75.4	81.0	6.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	137.5	81.5	74.9	131.3	80.4	73.7	126.0	79.1	41.6	84.4	40.3	82.8	39.1	81.2	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.3	13.2	13.8	92.7	4.3	2.7	10.7	94.6	8.2	96.1	5.8	97.6	2.7	99.5		
(5)			9.9	78.9	3.8	1.4	7.2	80.0	4.9	80.8	2.8	81.6	0.0	82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	34.1	36.8	44.1	54.0	61.9	67.4	74.4	75.7	67.7	59.5	48.8	38.3
	DBStd	15.30	6.03	6.24	7.00	7.08	6.33	5.01	5.69	4.76	4.27	5.15	6.51	6.62
	HDD50	1578	492	370	211	36	1	0	0	0	0	5	100	363
	HDD65	4415	957	789	648	338	139	29	2	0	20	180	487	826
	CDD50	3520	0	1	29	156	369	521	755	795	531	298	63	2
	CDD65	883	0	0	1	8	42	100	292	330	101	9	0	0
	CDH74	5216	0	0	1	33	193	367	2007	2311	295	9	0	0
	CDH80	1268	0	0	0	3	27	60	572	571	35	0	0	0
Wind	WSAvg	5.6	6.6	6.2	6.5	6.5	5.7	4.5	4.2	4.2	5.1	5.6	6.0	6.3
Precipitation	PrecAvg	50.6	1.7	1.7	2.4	3.0	3.1	4.8	8.9	9.9	8.0	3.1	2.6	1.3
	PrecMax	71.6	3.3	5.0	6.1	7.0	7.4	14.8	29.7	29.9	18.3	12.0	8.6	6.3
	PrecMin	33.3	0.2	0.2	0.5	0.5	0.4	0.7	2.3	2.2	1.7	0.8	0.3	0.2
	PrecStd	11.6	0.9	1.3	1.4	1.6	1.7	3.1	5.4	5.7	5.3	2.4	2.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	57.7	69.4	78.4	83.5	85.3	91.2	91.6	83.8	74.9	67.9	58.2
		MCWB	44.0	46.2	52.0	57.4	60.3	66.5	75.7	76.8	73.7	61.5	55.5	47.8
	2%	DB	48.5	52.7	63.2	72.7	78.8	81.3	88.2	88.0	79.9	71.9	64.5	54.3
		MCWB	39.1	42.8	48.8	55.2	59.7	66.7	74.9	76.4	70.7	61.4	54.1	45.2
	5%	DB	45.6	49.5	59.3	68.9	75.6	78.0	85.7	85.3	77.2	70.0	61.9	51.0
		MCWB	37.3	40.9	47.3	53.1	58.9	65.8	74.5	75.7	69.7	60.9	52.8	42.3
	10%	DB	43.2	46.7	55.5	65.3	72.5	75.3	83.2	83.2	75.0	67.9	59.4	48.4
		MCWB	35.5	38.6	45.0	52.1	57.7	65.9	73.9	75.3	68.2	60.0	51.0	40.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.2	48.6	54.3	62.1	65.5	72.1	79.1	79.8	75.5	67.2	60.1	50.6
		MCDB	49.9	53.5	64.3	72.2	74.8	77.5	85.0	85.7	82.0	70.2	64.2	56.3
	2%	WB	41.2	44.7	51.0	58.0	63.1	70.5	77.5	78.5	72.8	65.1	57.0	46.5
		MCDB	46.2	50.9	60.0	68.2	72.0	75.6	83.9	84.2	77.7	68.9	62.4	52.4
	5%	WB	38.9	42.0	48.5	55.9	61.7	69.2	76.2	77.4	71.1	63.5	54.5	43.7
		MCDB	43.7	48.1	56.1	64.7	70.0	74.1	82.8	83.0	75.5	67.8	59.7	49.4
	10%	WB	36.8	39.5	46.2	54.0	60.3	67.9	74.9	76.3	69.5	61.6	52.2	41.3
		MCDB	41.9	45.4	54.2	61.8	68.2	72.6	81.3	81.6	73.7	66.3	58.0	47.4
Mean Daily Temperature Range	5% DB	MDBR	12.3	12.5	13.0	13.5	12.5	9.6	8.5	9.3	11.0	13.4	13.2	12.8
		MCDBR	14.3	15.4	17.3	19.0	17.6	14.1	12.8	12.2	13.0	15.2	15.3	15.0
		MCWBR	10.2	11.1	10.9	10.4	8.6	6.2	5.4	5.7	8.2	9.8	11.0	11.2
	5% WB	MCDBR	12.5	14.4	15.5	15.7	14.3	9.9	10.3	10.6	11.5	12.2	12.8	13.8
		MCWBR	10.2	11.6	11.2	10.3	8.5	5.6	5.7	6.0	7.9	8.8	10.3	11.3
Clear-Sky Solar Irradiance	taub		0.416	0.477	0.571	0.608	0.614	0.661	0.623	0.560	0.489	0.478	0.444	0.407
	taud		1.992	1.872	1.708	1.687	1.719	1.685	1.830	1.990	2.119	2.015	2.029	2.045
	Ebn at Noon		229	229	220	222	224	213	220	232	242	227	218	222
	Edh at Noon		42	54	70	75	74	77	66	55	46	46	40	37
All-Sky Solar Radiation	RadAvg		753	996	1295	1545	1712	1624	1368	1353	1220	1069	767	681
	RadStd		62	97	127	117	165	141	193	167	145	96	98	56